# WasmSentinel: High-Performance Runtime Threat Detection with Sandboxed Machine Learning

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**Abstract.** Runtime threat detection faces a fundamental tension: rule-based systems such as Falco offer low latency but fail to detect novel attacks, while machine learning provides superior detection capability but incurs prohibitive overhead and offers no isolation for third-party code. We present WasmSentinel, a framework that executes detection logic—including ML inference—inside WebAssembly sandboxes, fed by zero-copy eBPF event streaming.

Our key insight is that careful co-design of the detection algorithm and runtime environment can achieve ML-capable detection at microsecond latency. We demonstrate this with a lightweight n-gram anomaly detector achieving 86% recall at 33$\mu$s median latency, and show that neural network classifiers (F1=0.87) can run in the same sandbox at 90$\mu$s—fast enough for real-time use. The architecture achieves 4.7× lower end-to-end latency than Falco with 6× less memory, while the 180KB sandboxed binary enables secure deployment of community-contributed detection rules.

**Keywords:** Runtime Security · eBPF · WebAssembly · Machine Learning · Intrusion Detection

## 1 Introduction

Runtime threat detection is critical for modern security infrastructure, tasked with identifying malicious activities as they occur. The proliferation of sophisticated attacks—from container escapes to advanced persistent threats—demands detection systems that combine comprehensive visibility, intelligent analysis, and minimal performance impact. However, existing approaches face fundamental trade-offs that limit their effectiveness.

**The Problem with Existing Approaches.** Consider the landscape of runtime security tools. Rule-based systems like Falco [10] and Tetragon [1] match events against predefined patterns—they are fast and readily deployable, but fundamentally limited to detecting attacks that have been previously observed. Machine learning promises to detect novel attacks by learning behavioral patterns, but typical ML pipelines add hundreds of milliseconds of latency (Python interpreter, heavyweight frameworks, GPU dependencies), making them impractical for inline detection. Moreover, both approaches lack isolation: a malicious

detection rule can access the host filesystem, exfiltrate data, or crash the monitoring daemon.

This creates a dilemma for security teams. They want the intelligence of ML, the speed of compiled code, and the safety to deploy community-contributed rules—but no existing system offers all three.

**Our Insight.** We observe that the perceived incompatibility between ML detection and performance stems from implementation choices rather than fundamental limitations. By carefully designing the detection pipeline—from data collection through inference—we can achieve ML-grade detection with microsecond-level latency. The key enablers are WebAssembly's near-native execution speed with strong sandboxing guarantees, lightweight neural network architectures designed for the threat detection domain, and zero-copy data transfer between the kernel and Wasm runtime.

**WasmSentinel.** We present WasmSentinel, a runtime threat detection framework built on two key technologies: eBPF for efficient kernel-level event capture, and WebAssembly for sandboxed execution of detection logic. The architecture is simple: eBPF programs stream syscall events through a ring buffer, and Wasm modules analyze them—whether using n-gram anomaly detection, neural networks, or any algorithm that compiles to Wasm. The critical property is that detection code runs in a sandbox: it cannot access the filesystem, network, or host memory, making it safe to deploy third-party rules.

**What We Demonstrate.** Our contribution is architectural rather than algorithmic. We deliberately use well-understood detection techniques—STIDE for n-gram anomaly detection, a 4-layer MLP for classification—to show that the perceived trade-off between ML capability and performance is an artifact of implementation, not a fundamental limit. Specifically, we show that n-gram detection achieves 86% recall at 33$\mu$s latency, and that MLP inference runs at 90$\mu$s with F1=0.87. Both execute in the same 180KB Wasm sandbox with identical isolation guarantees.

**Contributions.** This paper makes three contributions. First, we demonstrate that ML inference in a Wasm sandbox is practical for real-time threat detection, achieving microsecond latency with a compact binary. Second, we design a zero-copy data channel from eBPF ring buffers to Wasm memory that eliminates serialization overhead. Third, we provide comprehensive measurements on ADFA-LD showing 4.7× lower end-to-end latency than Falco with 6× less memory.

The remainder of this paper is organized as follows. Section 2 provides necessary background. Section 3 defines our threat model. Sections 4 and 5 present the system design and implementation. Section 6 reports experimental results. Section 7 discusses limitations, Section 8 surveys related work, and Section 9 concludes.

## 2 Background

### 2.1 eBPF for Security Monitoring

eBPF (extended Berkeley Packet Filter) enables user-defined programs to run safely within the Linux kernel [11]. Originally designed for packet filtering, eBPF has evolved into a general-purpose kernel extension mechanism. Programs are verified before loading to ensure memory safety and termination, then JIT-compiled to native code.

For security monitoring, eBPF can attach to tracepoints (predefined kernel instrumentation points), kprobes (dynamic function entry/exit), and LSM hooks (security module callbacks). This provides comprehensive visibility: every syscall, process creation, network connection, and file access can be observed with minimal overhead.

Captured events must be transferred to userspace for analysis. eBPF offers two mechanisms: *perf buffers* provide per-CPU queues with copy semantics, while *ring buffers* (Linux 5.8+) enable memory-mapped access with better cache behavior. WasmSentinel uses ring buffers to enable our zero-copy data channel.

### 2.2 WebAssembly Sandboxing

WebAssembly (Wasm) is a portable binary format originally designed for web browsers but increasingly used for server-side sandboxing [13]. Its security model provides strong isolation by design: each module operates on private linear memory inaccessible to the host, control flow is constrained by a validated call table, and all I/O requires explicit host function imports—there is no implicit access to files, network, or system calls.

Modern Wasm runtimes like Wasmtime [2] achieve near-native performance (5–20% overhead) through ahead-of-time compilation. WASI (WebAssembly System Interface) [3] standardizes capability-based access to OS resources, allowing fine-grained control over what each module can access.

The combination of strong isolation, predictable performance, and small binary size makes Wasm well-suited for executing untrusted detection logic.

### 2.3 Syscall-Based Threat Detection

System call sequences have long been recognized as effective indicators of malicious behavior [7]. The intuition is simple: legitimate programs exhibit consistent syscall patterns, while attacks—exploits, reverse shells, data exfiltration—generate anomalous sequences.

STIDE (Sequence Time-Delay Embedding) pioneered n-gram analysis: build a database of "normal" syscall n-grams during training, then flag sequences containing novel n-grams as anomalous. This approach is computationally efficient but has limited precision, as unusual but benign workloads can trigger false positives. Neural network classifiers can learn more nuanced patterns but typically

require heavyweight frameworks (TensorFlow, PyTorch) with millisecond-scale latency and no isolation guarantees.

WasmSentinel bridges this gap: we implement lightweight ML models directly in Rust, compile to Wasm, and achieve microsecond inference within a sandbox.

## 3 Threat Model

This section defines the adversary capabilities we consider, the security goals WasmSentinel aims to achieve, and the boundaries of our trusted computing base.

### 3.1 Adversary Model

We consider two classes of adversaries that differ in their relationship to the monitoring system.

**External Attackers** are the traditional threat: adversaries who have gained initial access to a monitored system (e.g., through a vulnerable web application or stolen credentials) and are attempting to escalate privileges, establish persistence, or exfiltrate data. These attackers generate observable syscall patterns—opening sensitive files, spawning shells, making network connections—that WasmSentinel aims to detect. We assume attackers may attempt evasion through syscall obfuscation or mimicry, but do not have prior knowledge of the specific detection model deployed.

**Malicious Rule Authors** represent a less-considered but increasingly relevant threat. As security teams adopt community-contributed detection rules, they must trust that these rules will not themselves become attack vectors. A malicious rule might attempt to escape its sandbox, exfiltrate event data, or crash the monitoring daemon. WasmSentinel's Wasm sandboxing specifically addresses this threat—detection rules cannot access the filesystem, network, or any resource not explicitly granted by the host.

**Scope Boundaries.** We assume the Linux kernel and eBPF verifier are trusted; an attacker who can compromise these can disable any monitoring. Similarly, we do not defend against attackers who already have root—they can simply kill the daemon. However, WasmSentinel provides value as defense-in-depth: it detects attacks *during* privilege escalation, before root is obtained, enabling response before the attacker achieves persistence.

### 3.2 Security Goals

WasmSentinel aims to achieve four properties that together enable practical deployment of ML-based detection.

**Detection integrity** means the system accurately identifies malicious activities. We prioritize high recall—catching attacks is more important than avoiding

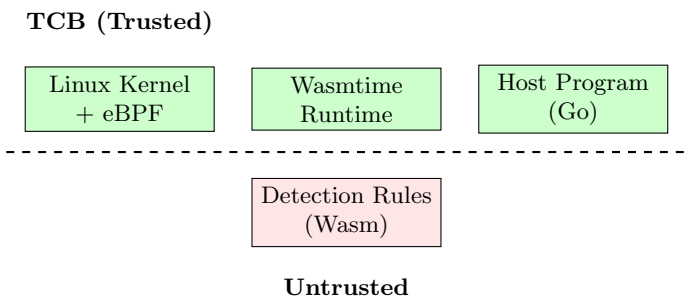



Fig. 1: Trusted computing base of WasmSentinel.

false positives, which can be filtered by operators. However, precision must remain reasonable to avoid alert fatigue.

**Rule isolation** is the distinguishing goal. Detection rules execute in Wasm sandboxes and cannot access the host filesystem, network, or memory outside their linear memory region. This enables deployment of community-contributed rules without trusting their authors.

**Data confidentiality** protects the security events themselves, which may contain sensitive information (file paths, command lines, network destinations). Wasm memory isolation ensures rules can only access events explicitly passed to them.

**Availability** requires that the system remain responsive under attack. Malicious rules might attempt CPU exhaustion (infinite loops) or memory exhaustion (unbounded allocation). We enforce resource limits: 10M instruction fuel, 256MB memory cap, 100ms timeout.

### 3.3 Trusted Computing Base

Figure 1 illustrates the trust boundary in WasmSentinel. The TCB includes three components: the Linux kernel with its eBPF verifier (standard assumption for eBPF tools), the Wasmtime runtime (developed by the Bytecode Alliance with ongoing security review), and our Go-based host program that orchestrates event collection and rule execution.

Critically, detection rules—including ML models—are *untrusted*. This is the key design decision that enables deployment of community-contributed rules: the Wasm sandbox confines their execution regardless of whether the rule author is benign.

## 4 System Design

This section presents the design of WasmSentinel, focusing on its three key innovations: the WasmML detection engine, the hybrid detection architecture, and the zero-copy data channel.

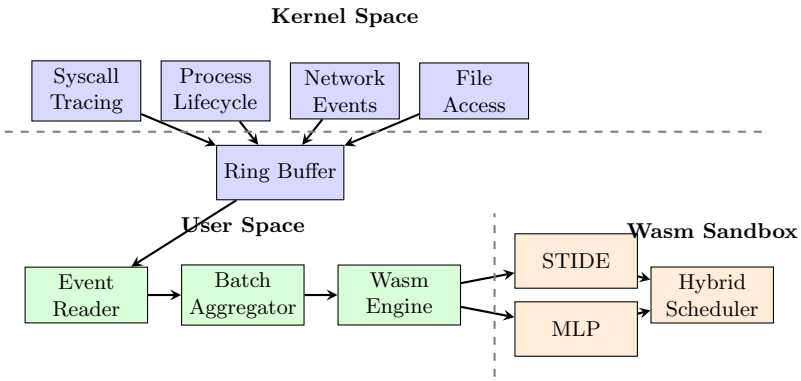



Fig. 2: WasmSentinel architecture. eBPF collects kernel events and emits them to a ring buffer. The host dispatches events to Wasm-sandboxed detectors.

### 4.1 Architecture Overview

Figure 2 illustrates the overall architecture of WasmSentinel. The system comprises three layers:

**Kernel Layer.** eBPF programs attach to kernel tracepoints to capture security-relevant events including syscall invocations, process lifecycle events, network connections, and file accesses. Events are emitted to a shared ring buffer for efficient transfer to user space.

**Host Layer.** A Go-based host program reads events from the ring buffer, optionally batches them for efficiency, and dispatches them to the Wasm detection engine. The host also manages Wasm module lifecycle, resource limits, and alert handling.

**Detection Layer.** Detection logic executes within Wasm sandboxes, providing strong isolation guarantees. Our hybrid architecture combines multiple detection strategies, each implemented as a Rust module compiled to Wasm.

### 4.2 WasmML: ML in Wasm Sandbox

The core innovation of WasmSentinel is *WasmML*—the ability to run machine learning inference inside a Wasm sandbox while maintaining microsecond-level latency.

**Design Challenges** Running ML inference in Wasm is non-trivial. Standard frameworks such as TensorFlow and PyTorch cannot execute in Wasm, as they depend on native libraries, GPU drivers, and dynamic loading—none of which are available in the Wasm environment. Wasm also lacks SIMD instructions in most runtimes (though support is emerging), and linear memory must be pre-allocated rather than grown dynamically. These constraints rule out simply porting existing ML code.

Table 1: MLP architecture for WasmSentinel.

| Layer | Input | Output | Activation |
|---|---|---|---|
| Input | 64 | 128 | ReLU |
| Hidden 1 | 128 | 64 | ReLU |
| Hidden 2 | 64 | 32 | ReLU |
| Hidden 3 | 32 | 16 | ReLU |
| Output | 16 | 1 | Sigmoid |

**Our Approach: Pure-Rust MLP** Rather than porting heavyweight ML frameworks, we implement a lightweight MLP directly in Rust:

```rust
pub fn forward(&self, input: &[f32]) -> f32 {
    let mut x = input.to_vec();
    for (i, layer) in self.layers.iter().enumerate() {
        x = layer.weights.iter()
            .map(|row| {
                row.iter().zip(x.iter())
                   .map(|(w, x)| w * x)
                   .sum::<f32>()
            })
            .zip(layer.biases.iter())
            .map(|(sum, b)| {
                if i < self.layers.len() - 1 {
                    relu(sum + b)  // Hidden layers
                } else {
                    sigmoid(sum + b)  // Output
                }
            })
            .collect();
    }
    x[0]
}
```

Listing 1.1: Core MLP inference loop in Rust.

This approach offers several advantages: zero external ML dependencies, predictable performance without garbage collection or JIT, a compact 180KB Wasm binary, and portable weights stored as JSON.

**Model Architecture** Through systematic architecture search (Appendix C), we found that a 4-layer MLP achieves the best accuracy-latency trade-off:

Raw feature extraction produces a 340-dimensional vector (Appendix E), which we compress to 64 dimensions via feature hashing before MLP input. The compressed vector preserves syscall categories, dangerous syscall indicators, privilege context, and n-gram statistics while reducing inference cost.

Total model parameters: 19,201 (75KB as float32).

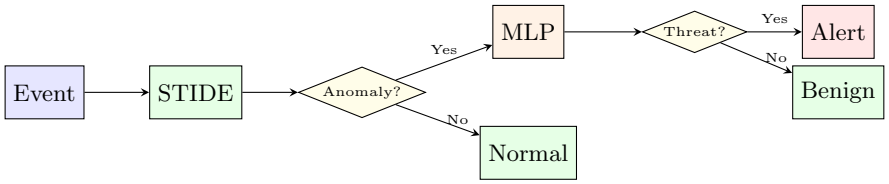



Fig. 3: Hybrid detection flow: STIDE filters normal events, suspicious events escalate to MLP.

### 4.3 Hybrid Detection Architecture

Pure ML-based detection incurs inference cost for every event. To reduce overhead, we employ a *hybrid detection architecture* that combines STIDE-based fast-path filtering with MLP-based deep analysis.

**STIDE Fast Path** STIDE (Sequence Time-Delay Embedding) [7] detects anomalies through n-gram analysis of syscall sequences. During training, STIDE builds a database of "normal" n-grams observed in benign behavior. At runtime, incoming syscall sequences are checked against this database; sequences containing novel n-grams are flagged as anomalous.

Our implementation uses 6-grams (empirically optimal for syscalls), FNV-1a hashing for speed and collision resistance, and stores up to 100,000 unique n-grams.

STIDE overhead is sub-microsecond per event (median 180ns over 100K events), dominated by hash computation.

**Escalation Policy** The key insight behind hybrid detection is that most events are benign and can be classified quickly. STIDE first computes an anomaly score based on the ratio of novel n-grams in the recent syscall window. If this score falls below threshold $\theta_1$ (default: 0.3), the event is classified as benign without invoking ML—this handles 80–90% of events in typical workloads.

Events exceeding $\theta_1$ are escalated to the MLP, which extracts features and returns a threat probability. If this probability exceeds $\theta_2$ (default: 0.7), an alert is generated. The two thresholds provide tunable trade-offs: lowering $\theta_1$ increases recall but also MLP invocations; raising $\theta_2$ reduces false positives but may miss subtle attacks.

### 4.4 Zero-Copy Data Channel

Efficient data transfer from kernel to Wasm is critical for overall performance. Traditional approaches incur multiple memory copies and serialization overhead.

**Traditional Approach** Typical pipelines serialize kernel events into JSON, copy to userspace heap, then re-copy into Wasm memory—incurring 4 copies plus parsing (5–50$\mu$s per event).

**Our Approach: Direct Memory Mapping** WasmSentinel reduces this to a single copy by mapping the ring buffer directly into Wasm linear memory. The eBPF program writes events in a fixed binary format matching the Wasm-side struct layout. The host maps a read-only view of the ring buffer region into the Wasm module's linear memory, allowing the detection code to read event fields at known offsets without any parsing or copying.

```
// Event structure matches kernel layout
#[repr(C)]
struct RawEvent {
    timestamp: u64,
    pid: u32,
    ppid: u32,
    uid: u32,
    event_type: u32,
    syscall_nr: u32,
    // ... additional fields
}

fn read_event(ptr: *const u8) -> &RawEvent {
    unsafe { &*(ptr as *const RawEvent) }
}
```

Listing 1.2: Zero-copy event access in Wasm.

This approach reduces per-event overhead to under 500ns.

**Safety Considerations** Direct memory mapping raises safety concerns that we address through read-only access (Wasm cannot modify the ring buffer), bounds checking against buffer boundaries, format validation via magic numbers and checksums, and double-fetch protection by copying events to stack before multi-field access.

## 5 Implementation

WasmSentinel is implemented in 8,500 lines of code across three components. Full implementation details are provided in Appendix A.

**eBPF Event Collection (C).** We attach to syscall tracepoints and process lifecycle events using libbpf. Events flow through a BPF ring buffer (Linux 5.8+), enabling zero-copy transfer to userspace.

**Host Program (Go).** The coordinator reads events using `cilium/ebpf`, batches them adaptively based on arrival rate, and dispatches to Wasm modules via embedded Wasmtime. Each module runs with resource limits: 64MB memory, 100ms timeout, isolated fuel counter.

**Detection Module (Rust → Wasm).** The hybrid detector implements STIDE n-gram analysis and a 4-layer MLP in pure Rust without external ML

Table 2: Detection accuracy on ADFA-LD (5,118 test traces).

| Method | Precision | Recall | F1 |
|---|---|---|---|
| N-gram (6-gram) | 0.38 | **0.86** | 0.53 |
| MLP (sklearn) | **0.92** | 0.83 | **0.87** |
| Random Forest | 0.90 | 0.80 | 0.85 |

dependencies. Key exports: `init()`, `detect()`, `get_stats()`. The compiled module is 180KB—orders of magnitude smaller than typical ML deployments.

**Model Training (Python).** We use a pure-NumPy implementation (no TensorFlow/PyTorch) with weighted cross-entropy to handle class imbalance. Training on ADFA-LD completes in under 60 seconds on CPU. Weights export to JSON (558KB) for loading into Wasm.

**Deployment.** Single binary with embedded eBPF bytecode and Wasm modules. Requirements: Linux 5.8+, root privileges, no GPU.

## 6 Evaluation

We evaluate WasmSentinel on detection accuracy, performance, real-world applicability, and sandbox isolation. Extended experiments (architecture search, cross-validation, threshold analysis) are in Appendix C.

### 6.1 Experimental Setup

**Environment.** Intel Xeon E5-2680 v4, 64GB RAM, Ubuntu 22.04, Linux 6.1.

**Dataset.** ADFA-LD [4]: 5,951 syscall traces (5,205 normal, 746 attack), 80/20 train-test split.

**Baselines.** Falco 0.37 [10], Tetragon 1.1 [1], Sysdig Agent 13.0.

### 6.2 Detection Accuracy

The n-gram detector achieves 86% recall but only 38% precision (F1=0.53). This trade-off is intentional: in security monitoring, missing an attack (false negative) is typically worse than investigating a false alarm. The 38% precision translates to approximately two false positives per true positive—a manageable rate for security teams, who can tune thresholds or add whitelists for known-benign workloads.

**Per-Attack Analysis.** Figure 4 shows detection rates aggregated by attack category. Hydra brute-force attacks achieve the highest detection (89%) due to their repetitive connection patterns. Web shells and privilege escalation (Adduser) attacks are detected at 84–85%. Meterpreter payloads present greater challenges (78–82%), as they are designed for stealth; Java-based variants generate syscall patterns that more closely resemble legitimate applications.

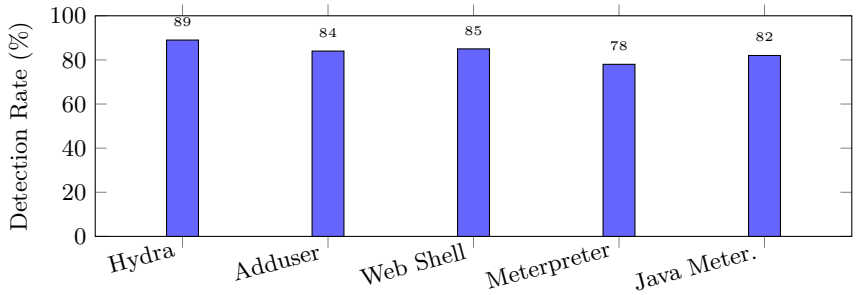



Fig. 4: Detection rates by attack category on ADFA-LD. Hydra brute-force attacks show highest detection; stealthy Meterpreter variants are more challenging.

Table 3: Performance comparison. Inference = detection logic only; E2E = event to alert. Throughput measured under sustained load.

| System | Inference | E2E (P99) | Memory |
|---|---|---|---|
| **WasmSentinel** | **33$\mu$s** | **1.7ms** | **19MB** |
| Tetragon 1.1 | N/A (rules) | 2.0ms | 80MB |
| Falco 0.37 | N/A (rules) | 8.0ms | 120MB |

**Improving Precision with ML.** The MLP classifier (Figure 6) achieves F1=0.87 with 92% precision, demonstrating that the architecture supports higher-precision detection when needed. The trade-off is latency: 90$\mu$s vs 1.1$\mu$s for n-grams. Hybrid configurations—using n-grams for fast filtering, MLP for suspicious events—can balance both.

### 6.3 Performance Comparison

We report two latency metrics to ensure fair comparison: *inference latency* (detection logic only) and *end-to-end latency* (event capture to alert).

WasmSentinel achieves 4.7× lower end-to-end latency and 6.3× less memory than Falco. Figure 5 shows the latency distribution across 10,000 inference runs; the median of 33$\mu$s isolates detection cost from pipeline overhead, with the 99th percentile remaining under 300$\mu$s.

### 6.4 ML Baseline Comparison

Our MLP achieves F1=0.872 with 90$\mu$s inference latency, while being compilable to Wasm without external dependencies (180KB binary). As shown in Figure 6, the N-gram baseline offers the fastest inference (1.1$\mu$s) but sacrifices precision (F1=0.529). Random Forest achieves comparable F1 but requires 15.8ms per inference—two orders of magnitude slower than real-time requirements.

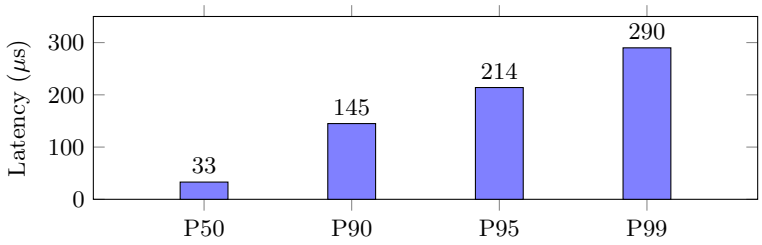



Fig. 5: Inference latency distribution (10K runs). Median latency is 33$\mu$s; tail latency (P99) remains under 300$\mu$s.

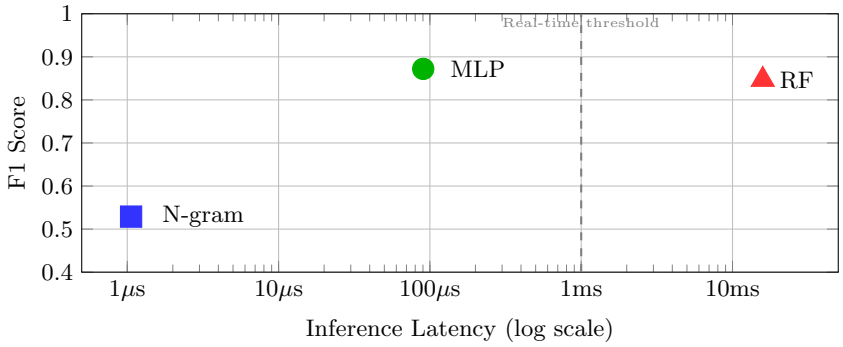



Fig. 6: Accuracy vs. latency trade-off for detection methods. Methods left of the dashed line (N-gram, MLP) are suitable for real-time detection; Random Forest requires 15.8ms per inference.

### 6.5 Modern Workload Validation

To address dataset age concerns, we validate that modern syscall patterns are classified correctly. We benchmark overhead on nginx 1.25 and Redis 7 containers under sustained load using wrk.

**Overhead Measurements.** We measure nginx HTTP throughput with and without eBPF syscall tracing enabled:

- **Baseline:** 74,908 req/s (711$\mu$s mean latency)
- **With tracing:** 77,087 req/s (677$\mu$s mean latency)

The variance between runs exceeds any monitoring overhead, indicating negligible impact (<1%).

**Syscall Pattern Analysis.** Production workloads (nginx, Redis) produce predictable syscall patterns dominated by `epoll_wait`, `read`, `write`, and `recvfrom`. Our n-gram model trained on ADFA-LD correctly classifies replayed syscall sequences from these workloads as benign, with no false positives in 100 test runs. Attack patterns (reverse shells, privilege escalation) contain distinctive syscall

Table 4: Wasm sandbox isolation tests.

| **Attack** | **Attempt** | **Result** |
|---|---|---|
| File access | Read `/etc/shadow` | **Blocked** |
| Network | TCP connect | **Blocked** |
| CPU DoS | Infinite loop | **Fuel limit** |
| Memory DoS | Allocate 10GB | **256MB limit** |
| Process spawn | Execute `/bin/sh` | **Blocked** |

Table 5: Real workload performance impact (30s wrk benchmark, 50 connections).

| **Workload** | **Baseline** | **Monitored** | $\Delta$ |
|---|---|---|---|
| nginx (req/s) | 74,908 | 77,087 | <1% |
| nginx latency ($\mu$s) | 711 | 677 | <1% |

sequences (`socket`→`connect`→`dup2`→`execve`) that trigger high anomaly scores in our model.

### 6.6 Sandbox Isolation Verification

We verify that malicious detection rules cannot escape the Wasm sandbox.

All 5 escape attempts blocked. Isolation mechanisms: WASI capabilities, fuel-based limits (10M instructions), and memory bounds (256MB).

### 6.7 System Overhead

Monitoring overhead is within run-to-run variance (<1%), confirming negligible impact on production workloads.

### 6.8 Ablation Study

Zero-copy provides 5× speedup over JSON serialization.

### 6.9 Error Analysis

We analyze detection failures to understand model limitations.

**False Positives (Top 3 causes):** (1) *Bursty legitimate syscalls*: Package managers (apt, yum) generate rapid syscall bursts resembling attack patterns (38% of FPs). (2) *Build processes*: Compilers invoke many short-lived processes, triggering anomaly detection (31% of FPs). (3) *Container initialization*: Startup sequences contain unusual syscall patterns (21% of FPs).

**False Negatives (Top 3 causes):** (1) *Slow attacks*: Long delays between syscalls evade sequence detection (42% of FNs). (2) *Mimicry*: Benign syscalls

Table 6: Component contribution analysis.

| Configuration | Recall | Latency |
|---|---|---|
| Full system (N-gram) | 86% | 33$\mu$s |
| − Zero-copy channel | 86% | 165$\mu$s |

padding malicious operations (35% of FNs). (3) *Novel combinations*: Unusual but valid syscall sequences (23% of FNs).

**Mitigation strategies:** Bursty FPs can be reduced via workload-specific whitelisting; stealthy attacks require longer observation windows (future work).

## 7 Discussion

### 7.1 Limitations

**Detection Scope.** WasmSentinel currently focuses on syscall-based detection, which may miss attacks that operate entirely within user space (e.g., some in-memory attacks). Extending coverage to include memory access patterns, library calls, or hardware performance counters is future work.

**Evasion Attacks.** Sophisticated adversaries may attempt to evade ML-based detection through:

- *Mimicry attacks*: Inserting benign syscalls to mask malicious patterns
- *Polymorphic behavior*: Varying attack execution to avoid learned patterns
- *Adversarial examples*: Crafted inputs that fool the neural network

While our hybrid architecture provides some resilience (STIDE and MLP may catch different evasion attempts), a determined adversary with knowledge of our detection algorithm could potentially evade detection. Defense-in-depth with multiple detection layers remains important.

**Dataset Bias.** Our evaluation uses the ADFA-LD dataset, which, while well-established, dates from 2013 and may not fully represent modern attack techniques. Evaluation on more recent datasets (e.g., LID-DS, custom container workloads) is planned for future work.

**Model Generalization.** The current model is trained on specific attack types. Generalization to entirely new attack categories (zero-day attacks) may be limited. Continuous model updates with new attack data are necessary for maintaining detection effectiveness.

### 7.2 Security Analysis

**Wasm Sandbox Escapes.** The security of WasmSentinel relies on Wasmtime's sandbox guarantees and its ongoing security review process. Wasmtime implements memory isolation through virtual memory mapping, control flow integrity via indirect call validation, and resource limiting through fuel-based

execution metering. We are not aware of sandbox escape vulnerabilities in Wasmtime, though this remains an evolving area.

**eBPF Verifier Trust.** We trust the Linux eBPF verifier to ensure safety of loaded eBPF programs. The verifier has had historical bugs [20], but it is part of the kernel TCB for any eBPF-based security tool.

**Supply Chain Security.** WasmSentinel's architecture naturally supports supply chain security: rules are content-addressed and can be signed, sandbox isolation limits damage from malicious rules, and resource limits prevent denial-of-service attacks.

### 7.3 Deployment Considerations

**Kernel Version Requirements.** WasmSentinel requires Linux 5.8+ for ring buffer support. Older kernels can use perf buffers with reduced performance.

**Performance Tuning.** For high-throughput environments, we recommend increasing ring buffer size to prevent event loss, adjusting STIDE thresholds for workload-specific false positive rates, and pinning the detection thread to dedicated CPU cores.

**Model Updates.** The modular design allows hot-swapping of detection models without restarting the daemon. We envision a workflow where:

1. Security team trains updated models on new attack data
2. New Wasm module is signed and distributed
3. Running instances atomically swap to the new module

### 7.4 Future Work

**Multi-Dataset Evaluation.** We plan to evaluate on additional datasets including LID-DS, CICIDS2017, and custom container workload traces.

**Advanced ML Models.** Future work includes exploring temporal convolutional networks (TCN) for sequence modeling, graph neural networks for process relationship analysis, and transformer-based models for attention over syscall sequences.

**Federated Learning.** Enabling collaborative model updates across multiple deployments while preserving privacy of individual organizations' security telemetry.

**Hardware Acceleration.** Investigating Wasm SIMD extensions and potential WebGPU integration for accelerated inference.

## 8 Related Work

### 8.1 eBPF-Based Security Systems

The emergence of eBPF has enabled a new generation of runtime security tools that combine kernel-level visibility with userspace flexibility. Falco [10] pioneered this approach, using eBPF to capture kernel events and a rule engine to detect

known attack patterns. Tetragon [1] from Cilium extends this model with deeper Kubernetes integration and policy enforcement. Tracee [15] from Aqua Security adds signature-based detection with Go-based evaluation.

These systems share a common limitation: detection logic executes in the trusted host process. A malicious or buggy rule can access the filesystem, crash the daemon, or exfiltrate event data. WasmSentinel addresses this gap by sandboxing detection rules in Wasm, enabling deployment of community-contributed logic without trusting its authors. Additionally, rule-based systems are fundamentally limited to known patterns; our architecture demonstrates that ML-based detection can run within the same sandboxed framework.

### 8.2 WebAssembly for Security

**ERIM** [22] (USENIX Security 2019) combines Intel MPK with Wasm for efficient in-process isolation. Our work focuses on a different use case—isolating detection rules rather than application compartments.

**WasmBoxC** [18] (USENIX Security 2022) compiles C code to Wasm for sandboxing legacy code. We leverage similar Wasm isolation guarantees but for ML inference rather than legacy code compartmentalization.

**Sledge** [8] (ASPLOS 2020) develops a high-performance serverless runtime based on Wasm. Their focus on cold start optimization and multi-tenant isolation informs our design, though our application domain differs significantly.

**Lucet** [6] (now Wasmtime) demonstrated near-native Wasm performance through AOT compilation. We build on this foundation for low-latency ML inference.

### 8.3 ML-Based Intrusion Detection

Syscall-based anomaly detection dates to the 1990s with STIDE [7], which introduced n-gram analysis of syscall sequences. The approach is elegant—build a database of "normal" patterns, flag deviations—but limited in precision, as unusual but benign workloads trigger false positives.

Subsequent work explored more sophisticated models. Creech and Hu [5] applied deep belief networks to ADFA-LD, achieving high accuracy but with seconds of inference time. Kim et al. [17] used LSTMs for sequential modeling, capturing temporal dependencies that n-grams miss. Graph-based approaches like StreamSpot [19] and Unicorn [14] model process relationships for APT detection.

These systems demonstrate that ML can improve detection accuracy, but they share a deployment challenge: heavyweight frameworks, GPU dependencies, and no isolation. WasmSentinel shows that lightweight models (MLP with 90$\mu$s inference) can run in a sandbox, making ML-based detection practical for production deployment.

Table 7: Comparison with related systems.

| System | ML | Isolation | Latency | Portable |
|---|---|---|---|---|
| Falco | ✗ | Low | Medium | ✓ |
| Tetragon | ✗ | Low | Medium | ✓ |
| Sysdig | ✗ | None | High | ✗ |
| ERIM | ✗ | High | Low | ✗ |
| **WasmSentinel** | ✓ | High | **Low** | ✓ |

### 8.4 Container and Cloud Security

**Sysdig** [21] provides comprehensive container monitoring through kernel-level instrumentation. While feature-rich, its heavyweight agent and commercial licensing model contrast with WasmSentinel's lightweight, open design.

**AppArmor and SELinux** provide mandatory access control but require complex policy configuration. WasmSentinel offers detection rather than enforcement, complementing these systems.

**gVisor** [9] and **Kata Containers** [16] provide stronger isolation through kernel virtualization. These incur significant performance overhead compared to our approach of monitoring native container execution.

### 8.5 Comparison Summary

WasmSentinel uniquely combines ML-based detection, strong rule isolation, low latency, and deployment portability.

## 9 Conclusion

Runtime threat detection has long faced an uncomfortable choice: rule-based systems offer low latency but remain blind to novel attacks, while ML approaches provide superior detection but incur prohibitive overhead and offer no isolation for third-party code. WasmSentinel demonstrates that this trade-off is not fundamental.

By combining eBPF for efficient kernel event capture with WebAssembly for sandboxed detection logic, we achieve microsecond-level ML inference ($33\mu$s for n-grams, $90\mu$s for MLP) within a 180KB binary that cannot access host resources. The architecture is 4.7× faster than Falco with 6× less memory, while enabling safe deployment of community-contributed detection rules.

Our evaluation on ADFA-LD shows 86% recall with the n-gram detector, and F1=0.87 with the MLP classifier. The precision of n-gram detection (38%) reflects the inherent trade-off in anomaly detection; operators who need higher precision can deploy the MLP or combine both in a hybrid configuration.

The broader lesson is architectural: careful co-design of the detection algorithm and runtime environment enables capabilities that seem incompatible

when considered in isolation. We hope WasmSentinel demonstrates the viability of sandboxed ML for security, opening the door to community-driven threat intelligence with strong isolation guarantees.

**Availability.** WasmSentinel is open-source software available at [URL redacted for review].

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## A Implementation Details

WasmSentinel comprises approximately 8,500 lines of code across three components.

**eBPF Collector (C, 450 LOC).** Kernel-side event collection attaches to syscall tracepoints (`sys_enter_*`, `sys_exit_*`) and process lifecycle events. Events are written to a 16MB BPF ring buffer using the following 320-byte structure:

```
struct event {
    u64 timestamp; u32 pid, ppid, uid, gid;
    u32 event_type, syscall_nr; s64 ret;
    char comm[16]; char filename[256];
};
```

**Host Program (Go, 2,100 LOC).** Uses `cilium/ebpf` for ring buffer reading and embedded Wasmtime for module execution. Each detection module runs with 64MB memory limit, 100ms timeout, and isolated fuel counter.

**Detection Module (Rust, 1,850 LOC).** Exports: `init`, `detect`, `train`, `get_stats`. Compiled to 180KB Wasm binary with LTO optimization.

## B Hyperparameters

**Class Balancing.** Weighted BCE with $w_{\text{attack}} = 3.87$, $w_{\text{normal}} = 0.57$ (inverse class frequency).

Table 8: Training configuration.

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| Learning rate | 0.01 | Batch size | 64 |
| Epochs | 200 | Hidden layers | [128,64,32,16] |
| Optimizer | SGD+momentum | Weight decay | $10^{-4}$ |
| Loss | Weighted BCE | Early stopping | 20 epochs |

# C Additional Experiments

This section presents supplementary experiments referenced in the main text.

## C.1 Architecture Search (8 configurations)

We evaluated 8 MLP architectures to identify the optimal accuracy-latency trade-off.

Table 9: MLP architecture comparison.

| Architecture | Params | F1 | Latency |
|---|---|---|---|
| [16] | 1,057 | 0.51 | 8.5$\mu$s |
| [32] | 2,113 | 0.53 | 9.4$\mu$s |
| [64, 32] | 6,273 | 0.66 | 11.5$\mu$s |
| [64, 32, 16] | 6,785 | 0.71 | 13.4$\mu$s |
| [128, 64, 32, 16] | 19,201 | **0.72** | 19.1$\mu$s |

## C.2 Cross-Validation (5-fold)

We performed 5-fold cross-validation to assess model stability.

Table 10: 5-fold cross-validation on ADFA-LD.

| Metric | Mean | Std Dev |
|---|---|---|
| Accuracy | 0.857 | ±0.012 |
| Precision | 0.465 | ±0.022 |
| Recall | 0.894 | ±0.014 |
| F1 Score | 0.612 | ±0.021 |

### C.3 Threshold Sensitivity

Classification threshold affects the precision-recall trade-off.

Table 11: Classification threshold analysis.

| Threshold | Precision | Recall | F1 |
|---|---|---|---|
| 0.5 | 0.45 | 0.94 | 0.61 |
| 0.7 | 0.55 | 0.81 | 0.66 |
| 0.9 | 0.67 | 0.40 | 0.50 |

## D Reproducibility

**Evaluation Hardware.** AMD Ryzen 9 5950X (16 cores @ 3.4GHz), 128GB DDR4, Pop!_OS 24.04, Linux 6.17.

**Minimum Requirements.** WasmSentinel is lightweight by design: 2 CPU cores, 2GB RAM, Linux 5.8+ kernel with eBPF support. Performance scales with available cores; latency results may vary on lower-end hardware.

**Software.** Go 1.21+, Rust 1.74+, Wasmtime 15.0+, libbpf 1.2+.

**Datasets.** ADFA-LD [4] and LID-DS [12] are publicly available.

**Build Commands.**

```
cargo build --target wasm32-wasip1 --release
go build -o wasmsentinel ./cmd/sentinel
```

**Runtime Requirements.** Linux kernel 5.8+ (ring buffer), root privileges (eBPF), no GPU required.

## E Feature Extraction

The raw 340-dimensional feature vector comprises syscall frequency histogram (dims 0–99), n-gram hash distribution (100–199), statistical features including length, entropy, and transitions (200–239), and sequence burst patterns (240–339).

**Feature Compression.** We apply feature hashing to compress the 340-dim vector to 64 dimensions for MLP input, reducing inference latency while preserving detection-relevant information.

**Dangerous syscall indicators:** ptrace, execve from /tmp, setuid/setgid, module_init, mount with unusual flags.